STAT 4310 Regression Project

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2026-04-19

class: center, middle, inverse

Section 1

STAT 4310 Regression Project

Part I: Two Regression Datasets

Part II: Finance Application

HousePrices, CASchools, and a 5-stock NYSE portfolio

Roadmap

- Part I uses two medium-sized datasets with optimal-model workflows.
- Part II uses five NYSE stocks from different industries plus the S&P 500 benchmark.
- Every section shows methods, interim process steps, diagnostics, extra validation, and final interpretation.

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Part I: Regression Methods

Step	Method
1. Describe data	response, predictors, missingness
2. Split Train/Test	80% / 20%
3. Build full model	lm()
4. Build CV ladder	nested backward models + repeated CV
5. Compare AIC vs CV	AIC-optimal vs CV-optimal
6. Remove outliers	Cook's distance
7. Check sensitivity	before vs after Cook filtering
8. Check residuals	QQ, BP, and DW tests
9. Bootstrap intervals	bootstrap coefficient CIs
10. Predict new cases	confidence and prediction intervals

HousePrices: Dataset Snapshot

.pull-left[- 546 observations and 12 variables - Response: `price` - `lotsize`: lot size of the house - `bedrooms`, `bathrooms`, `stories`, `garage`: size/layout features - `driveway`, `recreation`, `fullbase`, `gasheat`, `aircon`, `prefer`: yes/no housing amenities or location preference indicators - Missing values: 0 - The mix of structural and amenity variables makes the dataset useful for both interpretation and prediction.]

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Item	Value
Rows	546
Columns	12
Missing	0

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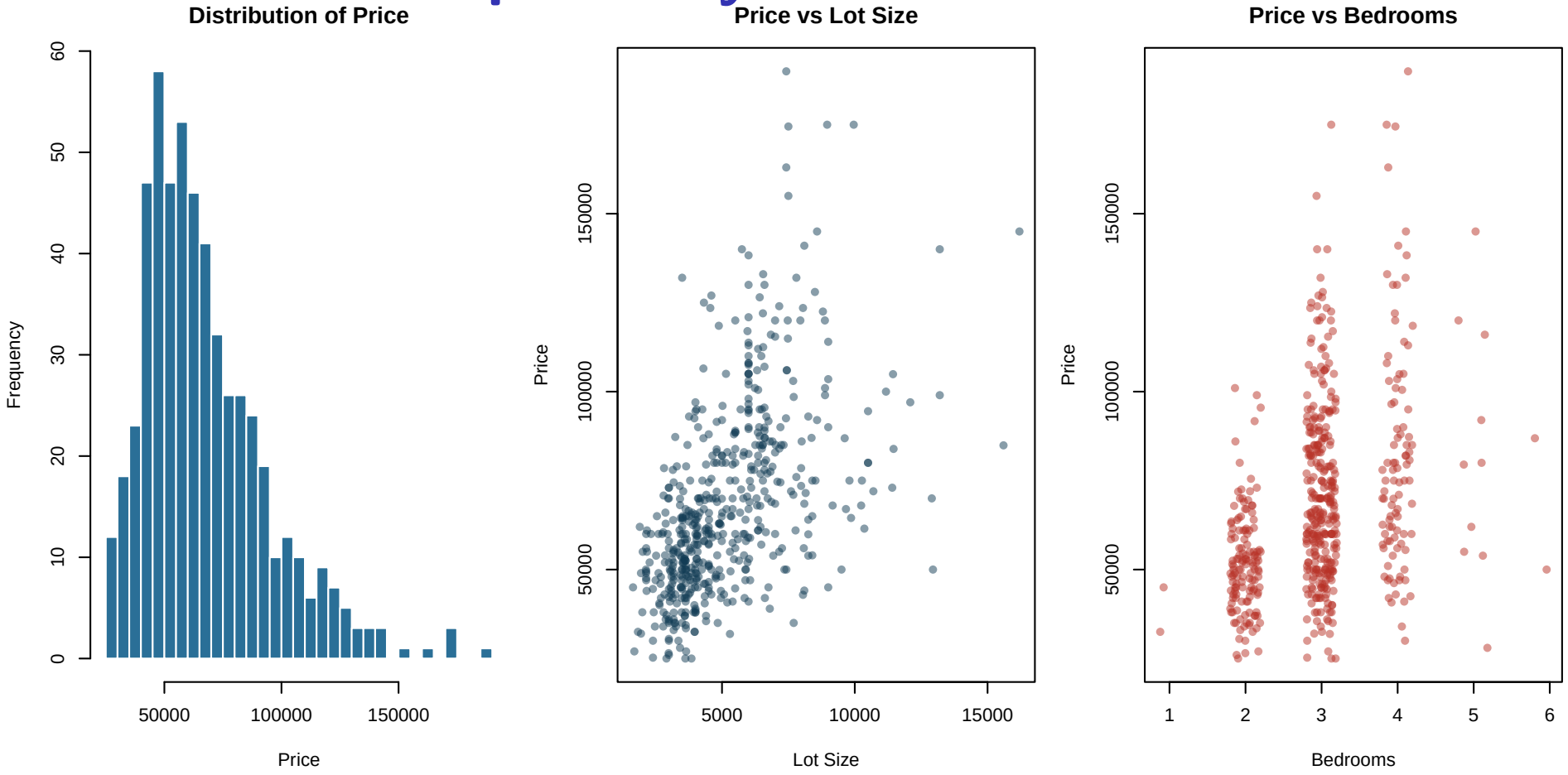
HousePrices: Exploratory Overview



- Price is right-skewed: most homes are modestly priced with a long tail of high-value sales.
- Lot size and price have a positive association, but with wide spread showing other features matter too.
- Bedroom count alone is a weak predictor — amenity combinations are

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HousePrices: Interim Process

Step	Detail
Split data	448 train / 98 test
Build full model	11 starting predictors
Build model ladder	11 nested steps
Run repeated CV	5 folds x 6 repeats
Compare AIC and CV	Step 0 vs Step 1
Remove outliers	36 points above Cook cutoff
Check residuals	QQ, BP, and DW tests
Bootstrap intervals	200 bootstrap resamples
Predict new homes	2 cases with CI and PI

- The ordering matters: choose the model on training data, check influential cases, then touch the held-out test set only once.

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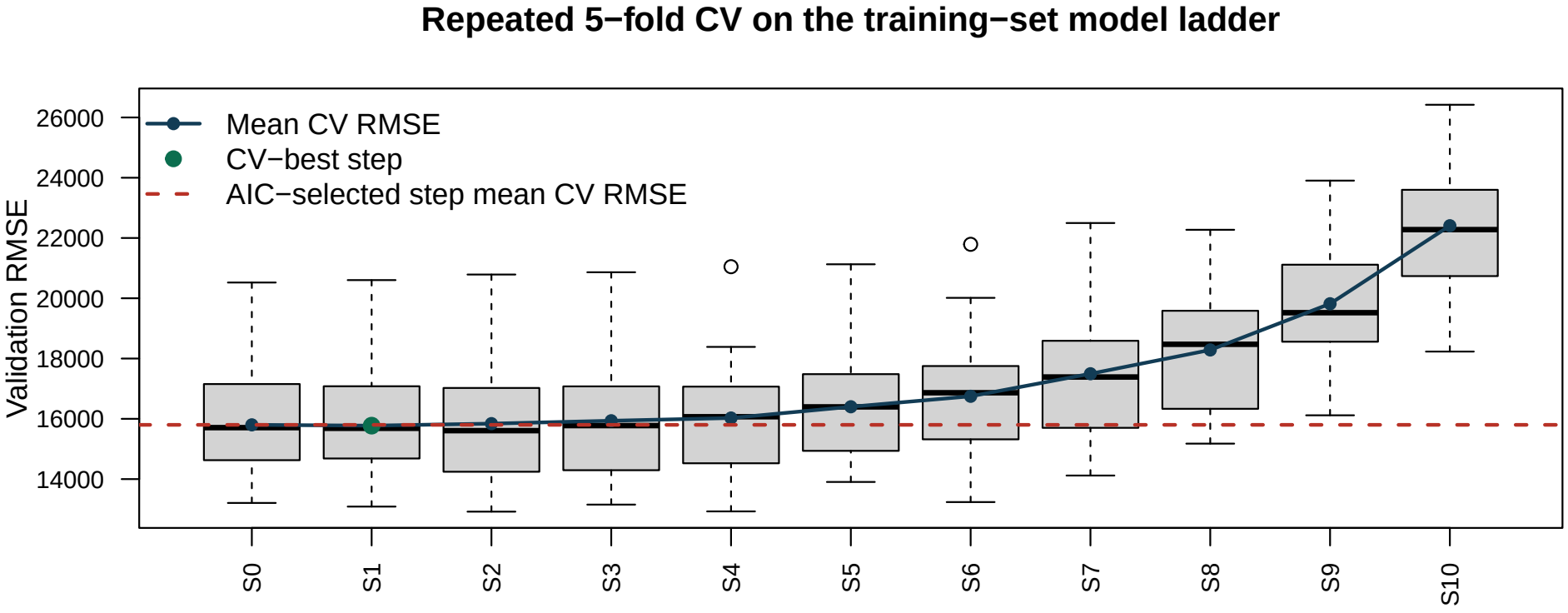
HousePrices: CV Across the Selection Ladder



- S0 is the full model.
- Each later step removes one more predictor from a nested ladder.
- Each step shows 30 validation RMSE values because we run 5 folds and repeat that 6 times.
- The red dashed line marks the AIC-selected step's mean CV RMSE.
- The green point marks the true CV minimum, which is S1 here.
- One early predictor removal improves validation RMSE, but additional

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HousePrices: AIC Step vs CV Step

Model	Best Step	Predictors	CV RMSE
AIC-optimal	Step 0	11	15800.09
CV-optimal	Step 1	10	15773.42

- AIC and CV are compared on the same training-set ladder.
- The held-out 20% test set is used only for the final check.
- For `HousePrices`, the CV gap is only 26.66, so the test flip of 234.64 is normal one-split noise rather than a contradiction.
- In practice, both criteria are pointing to nearly the same part of the ladder rather than arguing for very different housing models.

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HousePrices: Sensitivity to Cook Filtering

Training Data Used	AIC Best Step	AIC RMSE	CV Best Step	CV RMSE
Original training data	Step 0	15800.09	Step 1	15773.42
Cook-filtered training data	Step 1	11525.72	Step 1	11525.72

- The upper table asks whether influential sales change model choice or mostly change the size of the validation error.

Training Data Used	AIC Best Step	CV Best Step	Held-out Test RMSE
Original training data	Step 0	Step 1	AIC 15361.11, CV 15595.75
Cook-filtered training data	Step 1	Step 1	15195.88 (same for AIC and CV)

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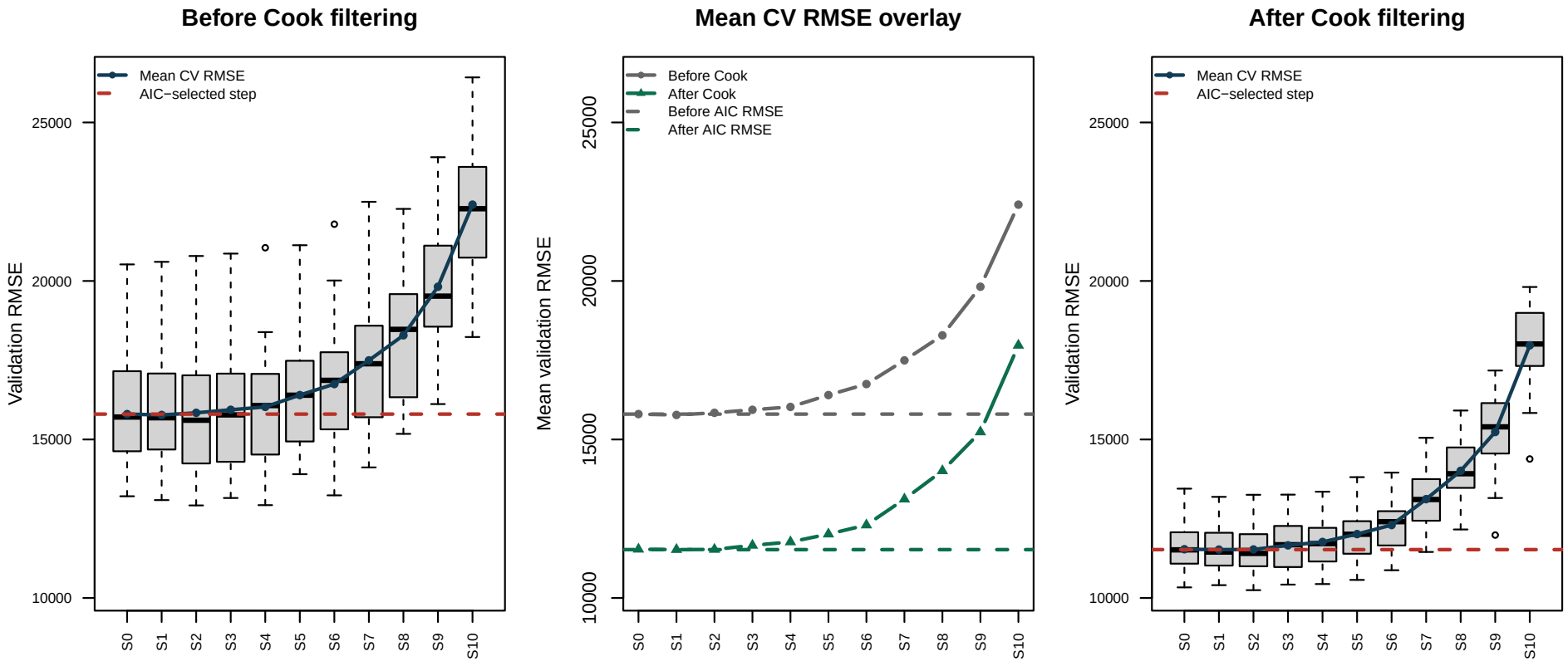
HousePrices: CV Ladder Before and After Cook Filtering



- The whole ladder shifts down after filtering, so the influential cases were hurting validation performance across multiple candidate models.
- Because the whole curve moves down, Cook filtering improves model stability rather than just rescuing one lucky final step.

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HousePrices: Final Model Summary

- Final predictors kept:

lotsize, bedrooms, bathrooms, stories, driveway, recreation, fullbase, gasheat, aircon, garage, prefer

- Removed from full model:

None from the full model

- The final model keeps the variables that most directly describe usable house characteristics and preferred location.
- Because no term drops from the full model, the main housing features appear to add complementary signal rather than redundant overlap.

Metric	Value
Adjusted R2	0.7418
Residual s	11281.6100
Outliers Removed	36.0000
Test RMSE	15290.8400
Test R2	0.6610

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HousePrices: Final Model Terms

Term	Estimate	P Value
lotsize	3.7246	$<$ 1e-16
stories	7513.8128	$<$ 1e-16
bathrooms	11862.8341	1.10e-16
airconyes	11024.1191	1.58e-15
preferyes	8693.1604	5.30e-09
garage	2990.4355	5.96e-05
fullbaseyes	5195.2531	0.000154
gasheatyes	11673.0625	0.000359
drivewayyes	5494.4737	0.001009
recreationyes	2702.0028	0.097389
bedrooms	1115.7889	0.213757

- Larger lots, more stories and bathrooms, air conditioning, a garage, and preferred location all move fitted price upward when the other features are held fixed.
- The coefficient table shows that the strongest evidence sits in structural

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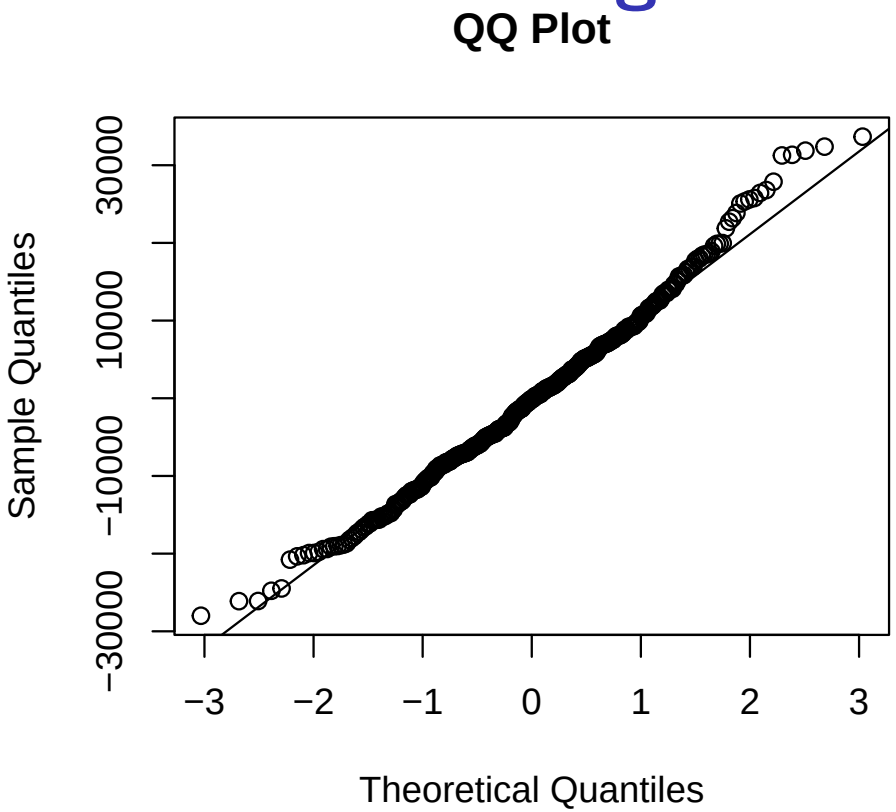
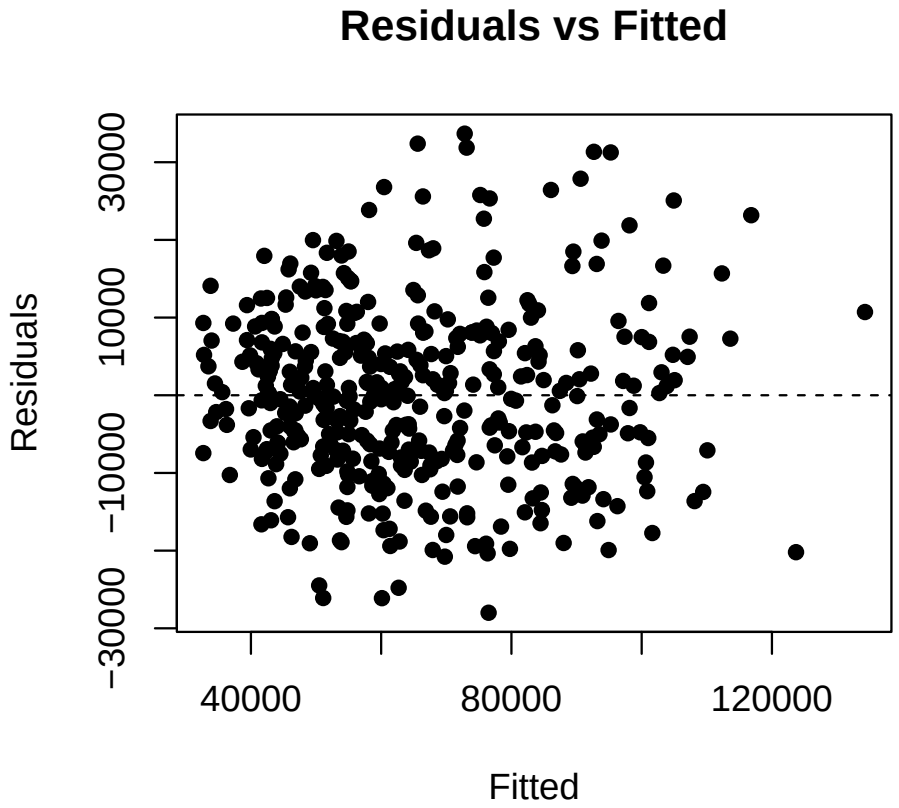
HousePrices: Diagnostics





Test	P.Value
Shapiro-Wilk	0.021159
Breusch-Pagan	0.009426
Durbin-Watson	0.000158

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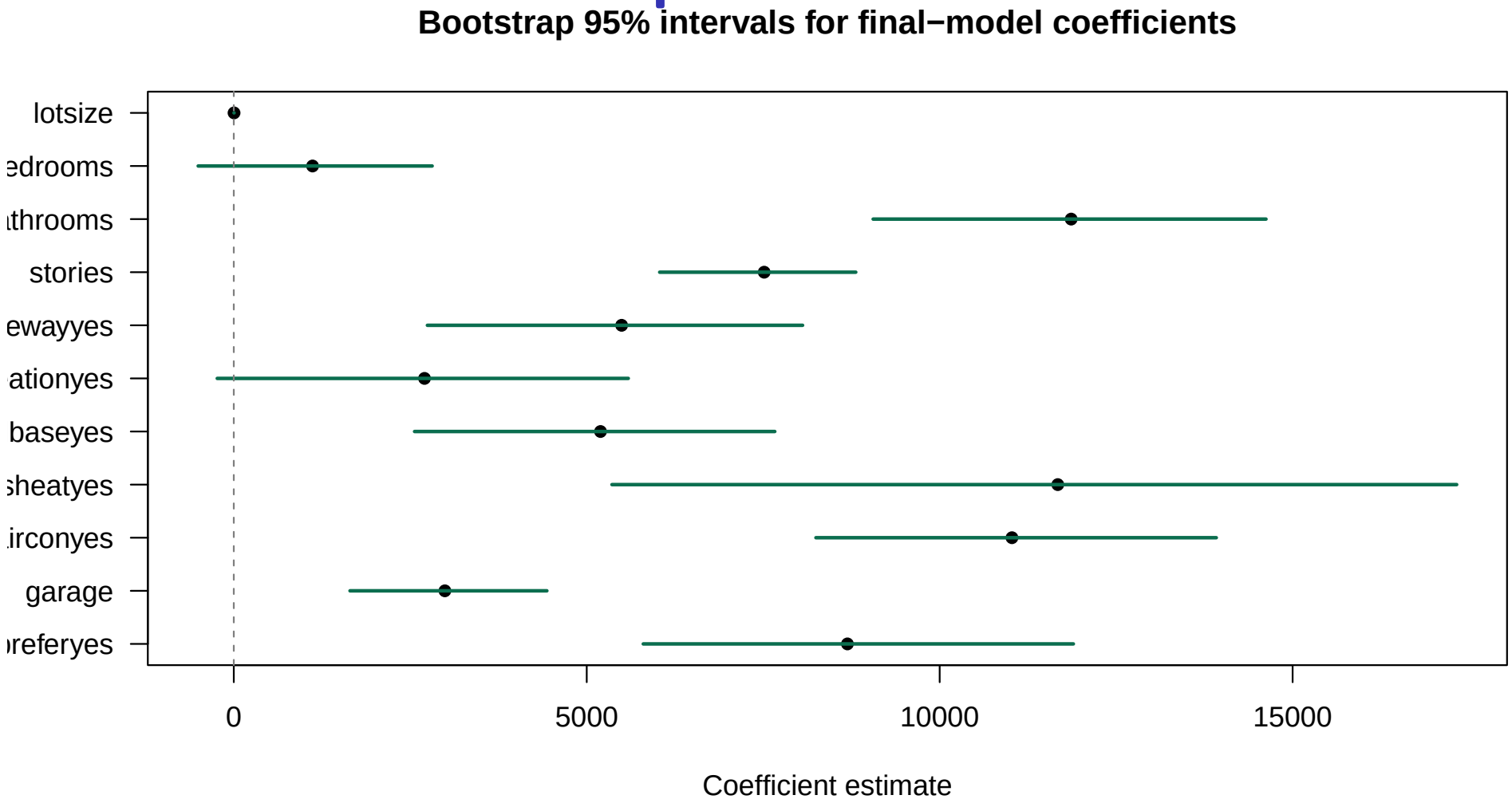
HousePrices: Bootstrap Coefficient Intervals



- These are empirical 95% intervals from 200 bootstrap resamples of the final training set.
- Intervals far from 0 support stable coefficient direction.
- The intercept is omitted because its much larger scale would compress

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HousePrices: Prediction Examples

Scenario	Fit	CI	PI
Smaller home	47871	(45465, 50277)	(25562, 70180)
Larger home	124806	(117441, 132170)	(101436, 148175)

- The confidence interval estimates the mean price.
- The prediction interval is wider because it predicts an individual home sale.
- The higher-amenity scenario also gets the higher fitted price, which matches the coefficient signs in the final model.

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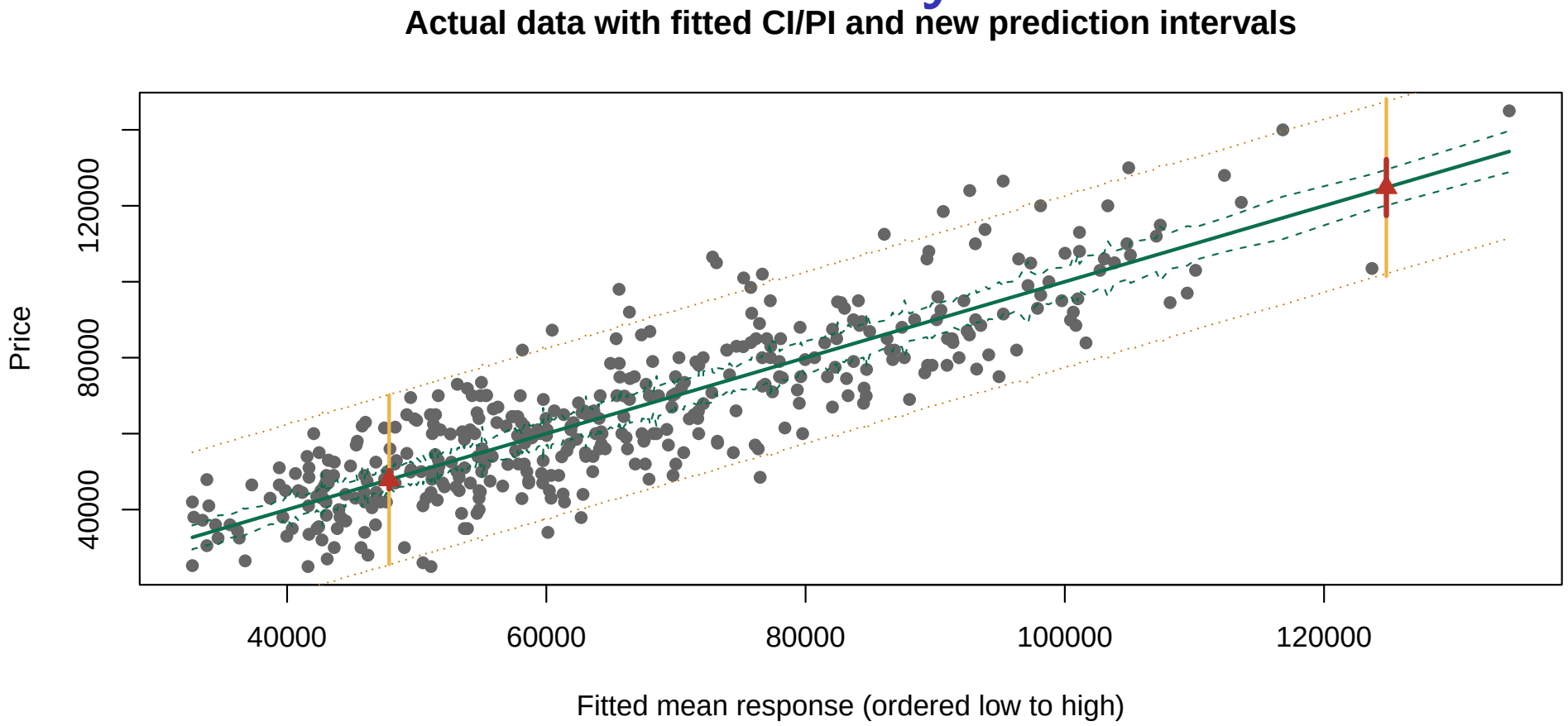
HousePrices: CI and PI Overlay



- Gray points are actual observed prices.
- The x-axis is fitted mean response, sorted from low to high.
- The green line is the fitted mean.
- Orange bands show PI for observed-data rows.
- Red triangles and bars now sit at their fitted locations on the green line.

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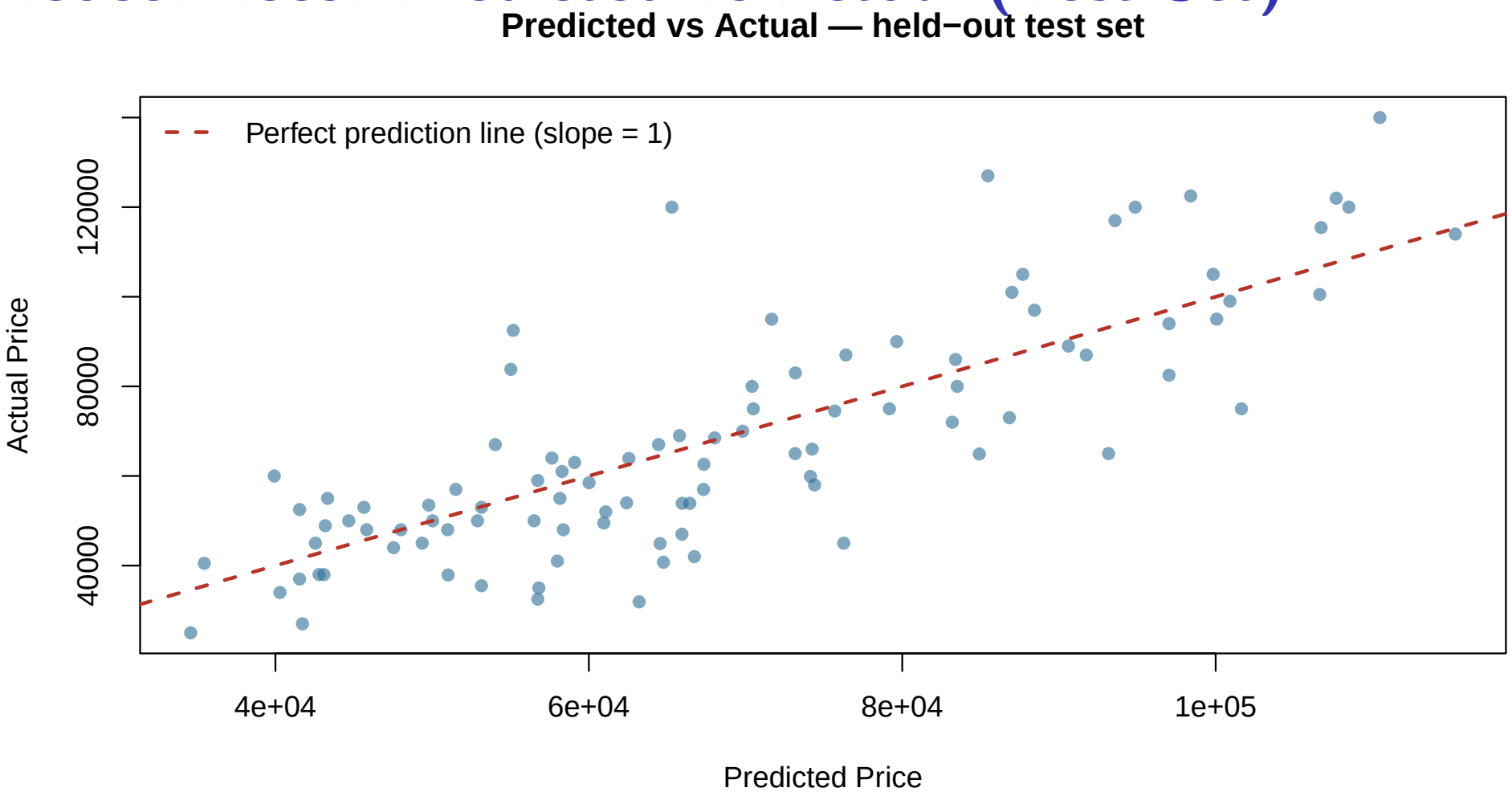
HousePrices: Predicted vs Actual (Test Set)



- Points close to the dashed line indicate accurate test-set predictions.
- Deviations are largest at high price values, showing more uncertainty for expensive homes.
- The overall scatter is consistent with the test RMSE reported in the

CASchools: Dataset Snapshot

.pull-left[- 420 observations and 9 analysis variables - Response: average test score `score` - `str`: students per teacher - `calworks`: percent of students on CalWORKs - `lunch`: percent eligible for subsidized lunch - `income`: district income level - `english`: percent of English learners - `grades`: district grade-span category; one fitted indicator is `gradesKK-08` for K-8 districts - Missing values: `0` - The covariates mix school resources with student-composition measures, so the model is mainly explanatory rather than causal.]

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Item	Value
Rows	420
Columns	9
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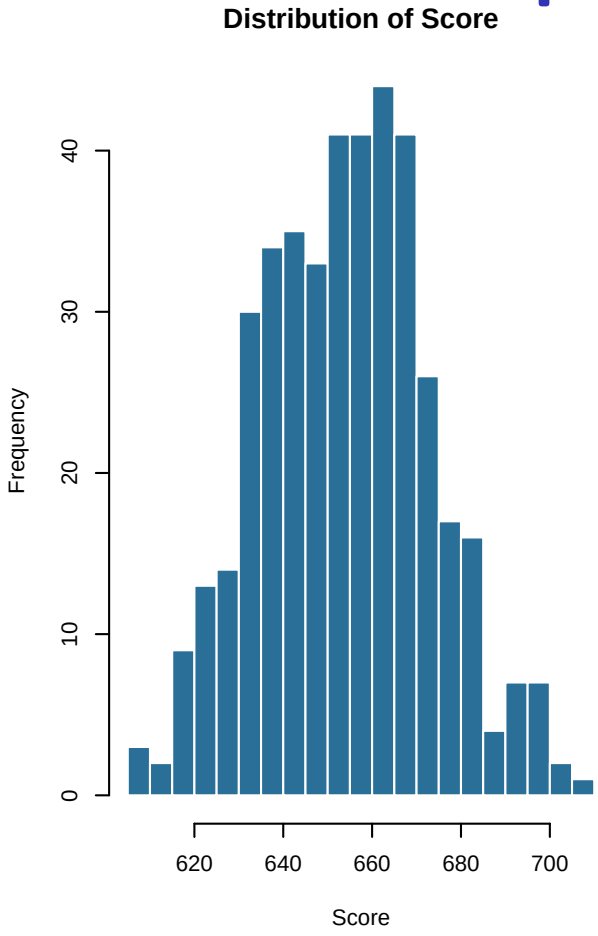
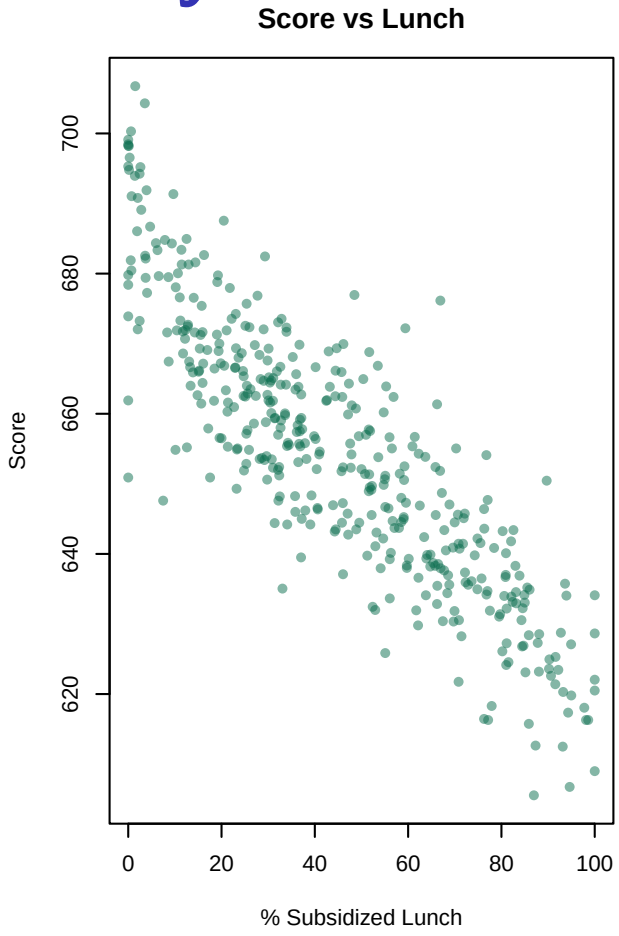
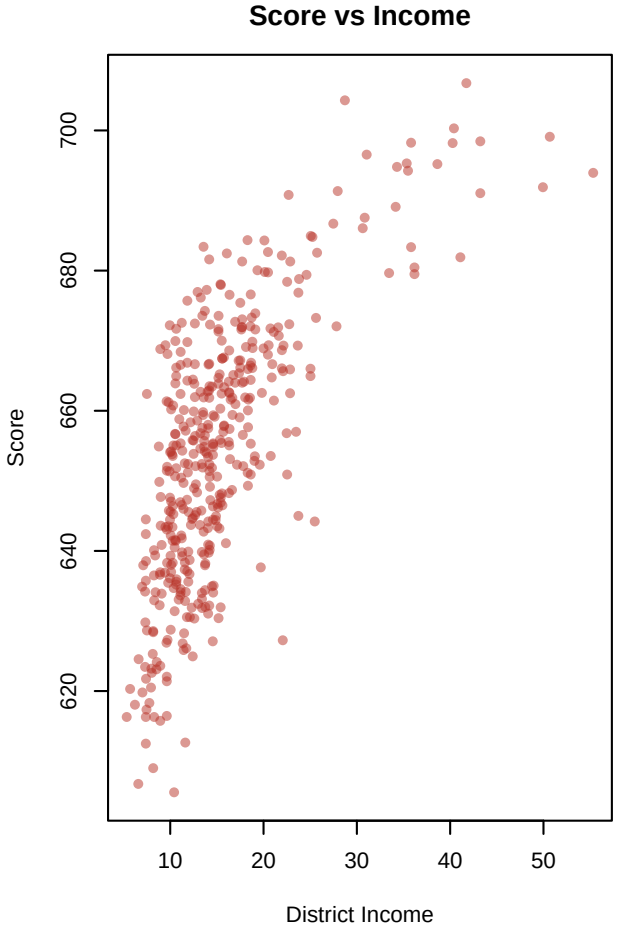
CASchools: Exploratory Overview







- Score is roughly symmetric and bell-shaped across districts.
- The negative relationship with lunch rate is the strongest raw signal in the data.
- Income shows a clear positive association, confirming that socioeconomic factors dominate the score pattern.

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CASchools: Interim Process

Step	Detail
Split data	326 train / 94 test
Build full model	8 starting predictors
Build model ladder	8 nested steps
Run repeated CV	5 folds x 6 repeats
Compare AIC and CV	Step 3 vs Step 3
Remove outliers	18 points above Cook cutoff
Check residuals	QQ, BP, and DW tests
Bootstrap intervals	200 bootstrap resamples
Predict districts	2 cases with CI and PI

- The main question is whether school-score variation is driven more by student composition than by resource variables.

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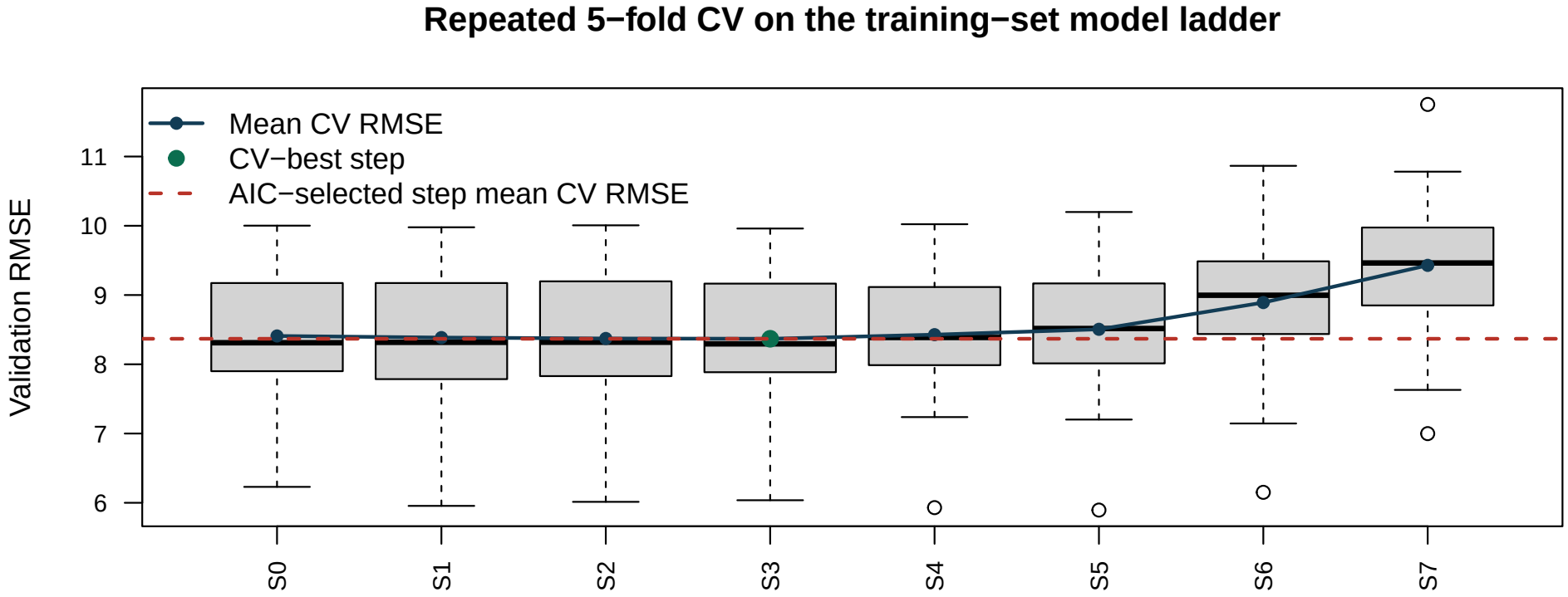
CASchools: CV Across the Selection Ladder



- The ladder is built from the training-set full model.
- Each step again summarizes 30 validation RMSE values from 5 folds repeated 6 times.
- This checks whether a simpler school-score model is competitive before touching the test set.
- Here the AIC step and CV step agree, so the red line lands at the best step.
- Agreement between AIC and CV makes the reduced school model

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CASchools: AIC Step vs CV Step

Model	Best Step	Predictors	CV RMSE
AIC-optimal	Step 3	5	8.3683
CV-optimal	Step 3	5	8.3683

- The agreement here says the AIC-selected reduction is not just an artifact of one criterion.

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CASchools: Sensitivity to Cook Filtering

Training Data Used	AIC Best Step	AIC RMSE	CV Best Step	CV RMSE
Original training data	Step 3	8.37	Step 3	8.37
Cook-filtered training data	Step 3	7.09	Step 3	7.09

- The main question is whether influential districts materially change the selected CV step or the full validation curve.

Training Data Used	AIC Best Step	CV Best Step	Held-out Test RMSE
Original training data	Step 3	Step 3	8.54 (same for AIC and CV)
Cook-filtered training data	Step 3	Step 3	8.68 (same for AIC and CV)

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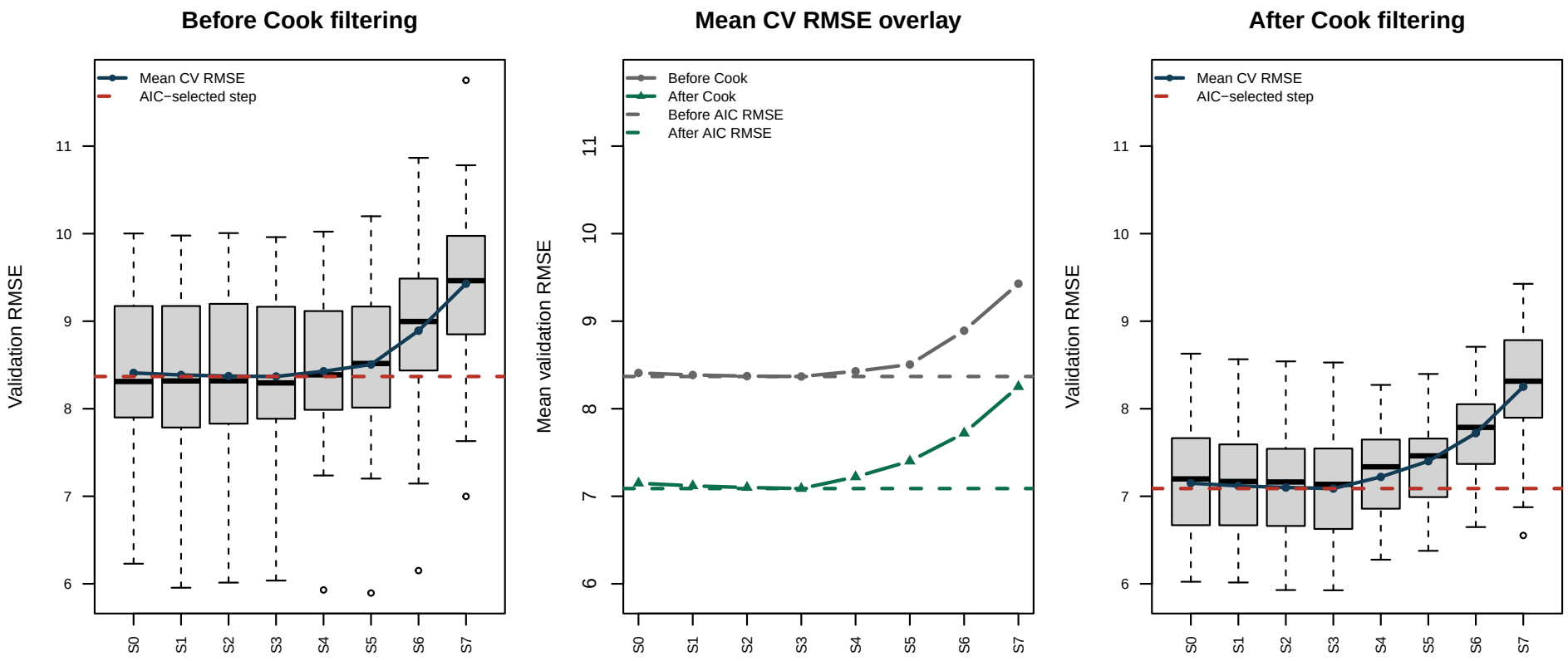
CASchools: CV Ladder Before and After Cook Filtering



- The best step stays the same after Cook filtering, but the full curve shifts down, which makes the selected model look more stable.
- That means filtering helps fit quality, but it does not change which predictors matter.

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CASchools: Final Model

- Final predictors kept:

str, lunch, income, english, grades

- Removed from full model:

calworks, computer.rate, expenditure

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Metric	Value
Adjusted R2	0.8560
Residual s	7.0500
Outliers Removed	18.0000
Test RMSE	8.6800
Test R2	0.7701

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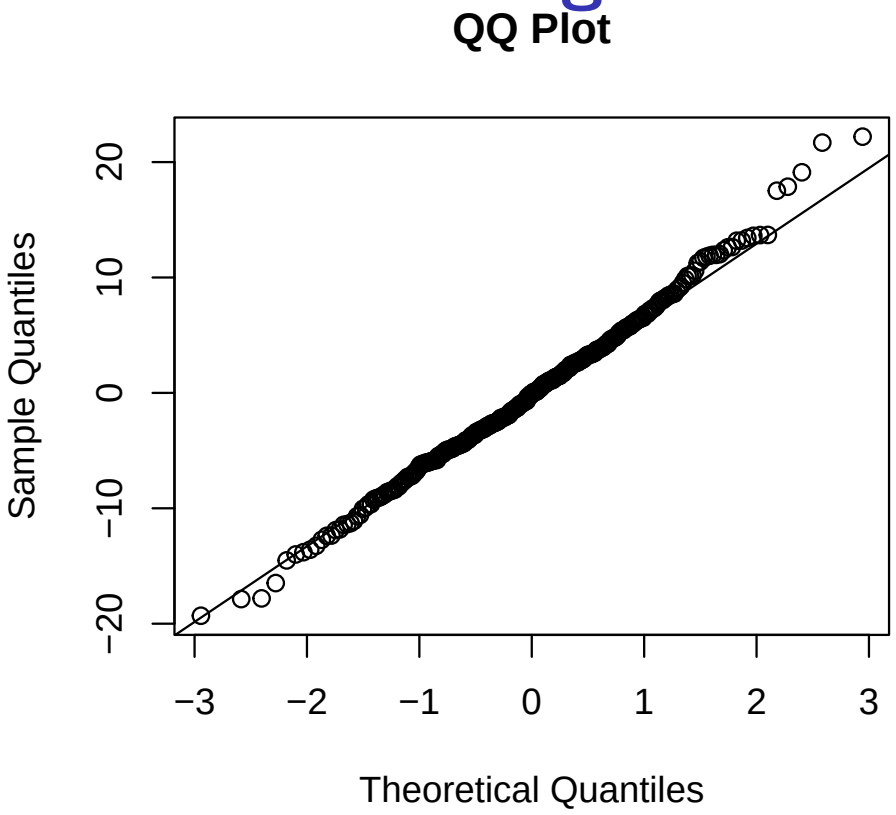
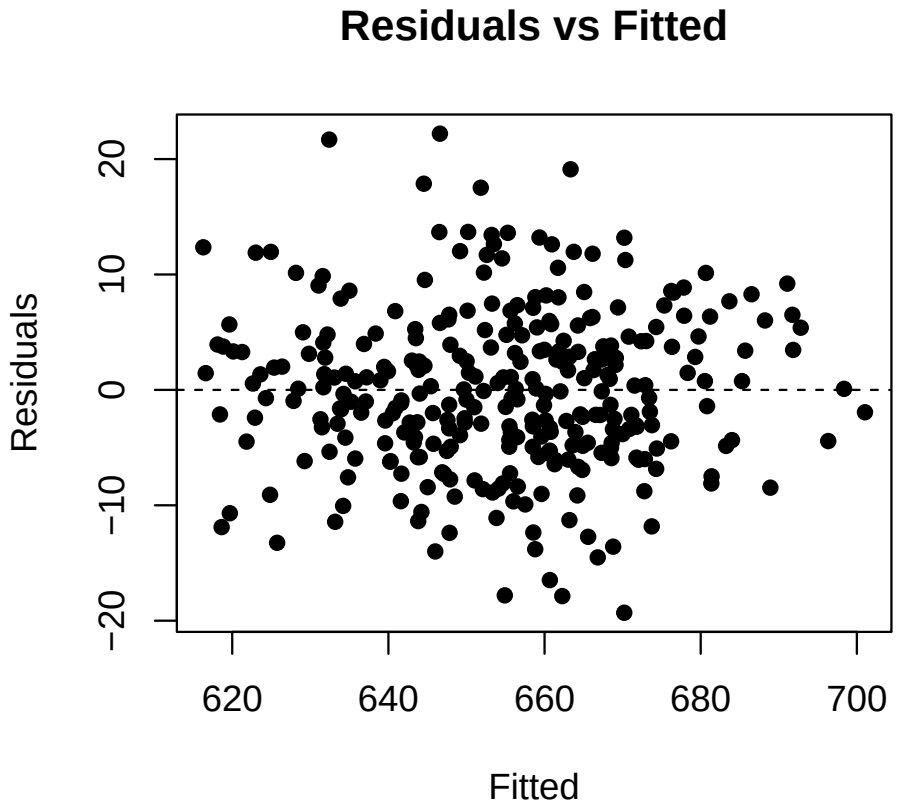
CASchools: Diagnostics





Test	P.Value
Shapiro-Wilk	0.5591
Breusch-Pagan	0.0474
Durbin-Watson	1.39e-05

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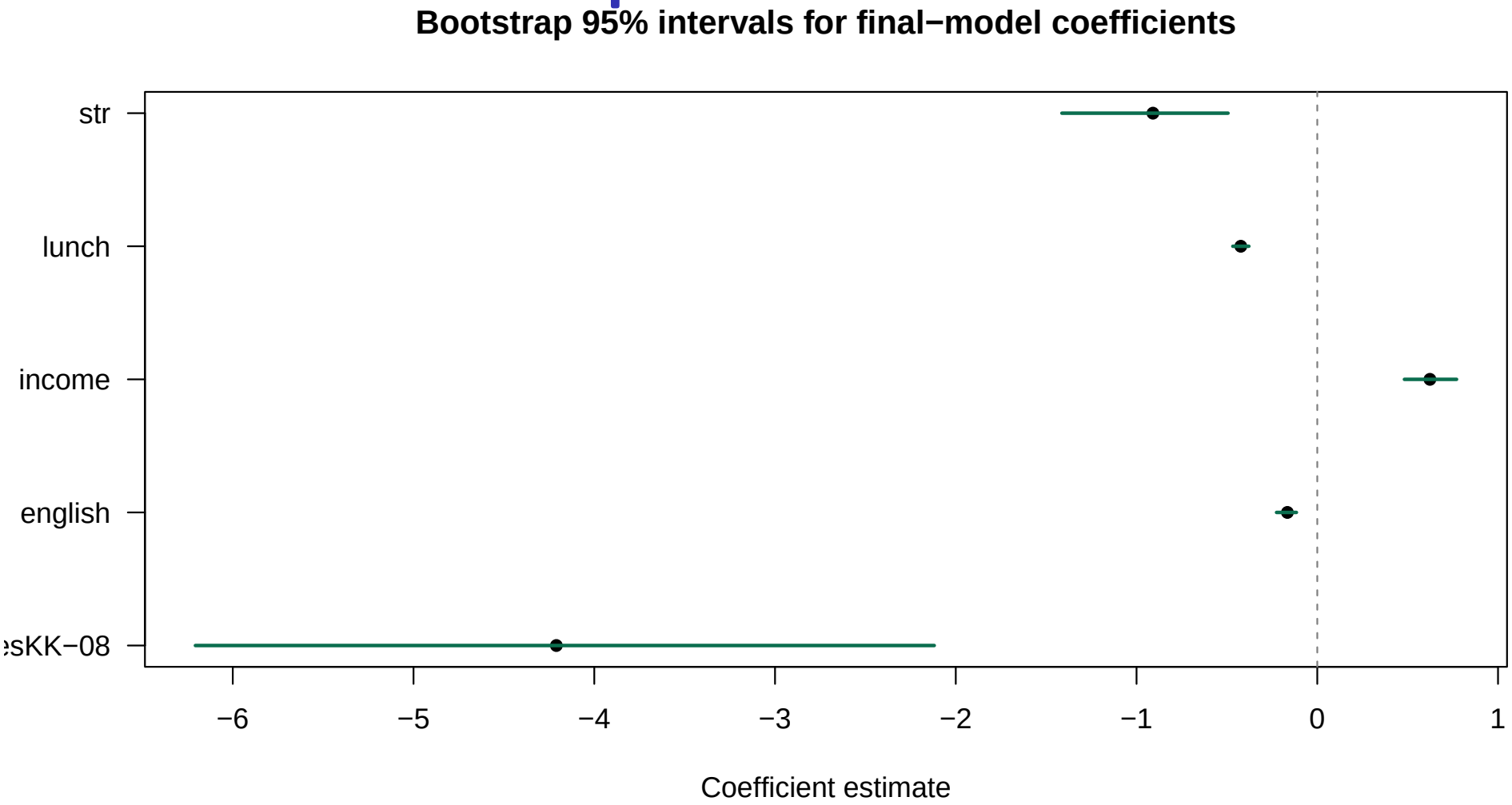
CASchools: Bootstrap Coefficient Intervals



- These are empirical 95% intervals from 200 bootstrap resamples.
- A log scale would not work here because several coefficients are negative.
- The intercept is omitted so the predictor intervals do not collapse

class: small-table

CASchools: Prediction Examples

Scenario	Fit	CI	PI
Lower-risk district	672.56	(670.3, 674.82)	(658.5, 686.62)
Higher-risk district	636.56	(635.06, 638.06)	(622.6, 650.52)

- Lower lunch rates and higher district income lead to higher predicted scores in the final model.
- The wider prediction intervals again show that district-level mean structure is estimated more tightly than individual-district outcome.

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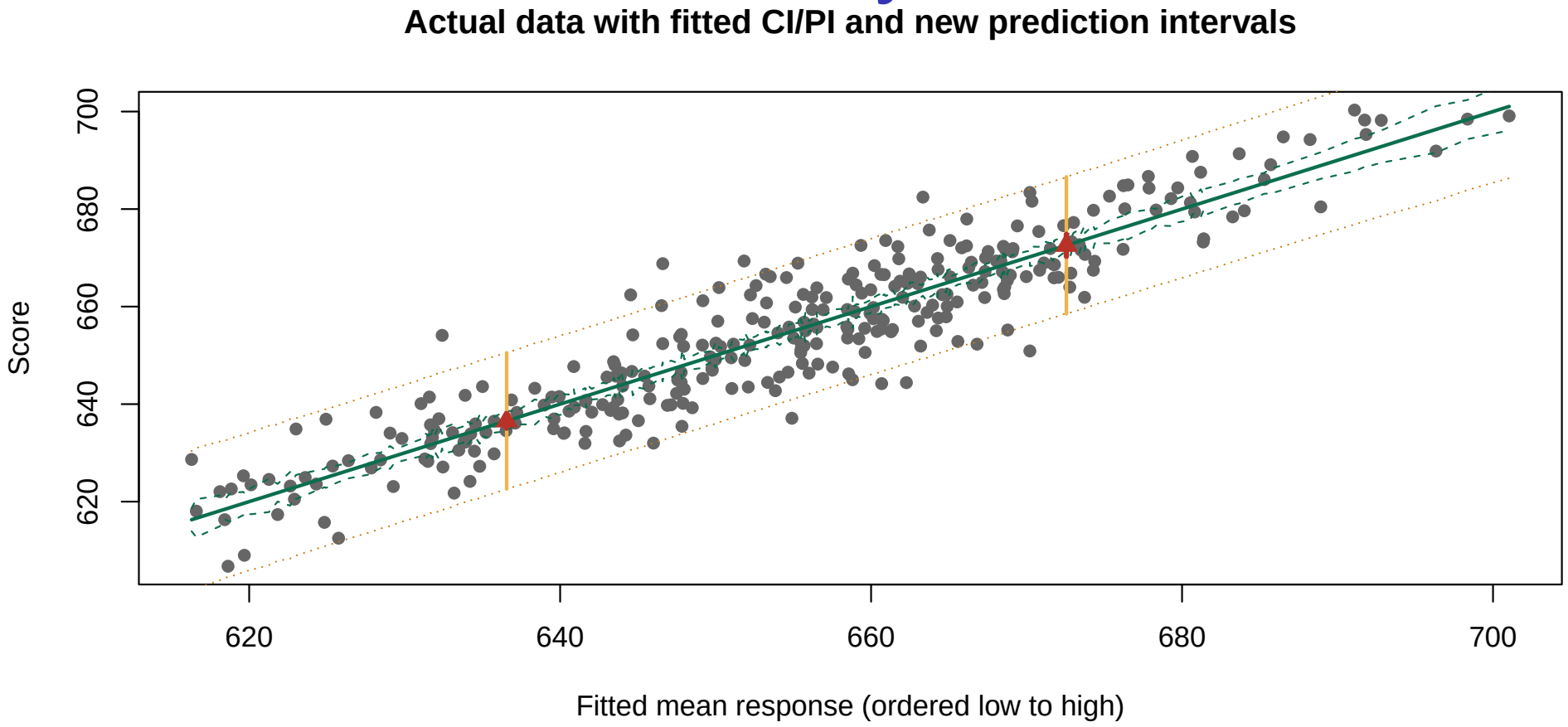
CASchools: CI and PI Overlay



- The x-axis is fitted mean response, sorted from low to high.
- New prediction cases now sit at their own fitted locations instead of being appended at the end.
- The spread around the fitted line is consistent with the wider orange prediction bands for new districts.

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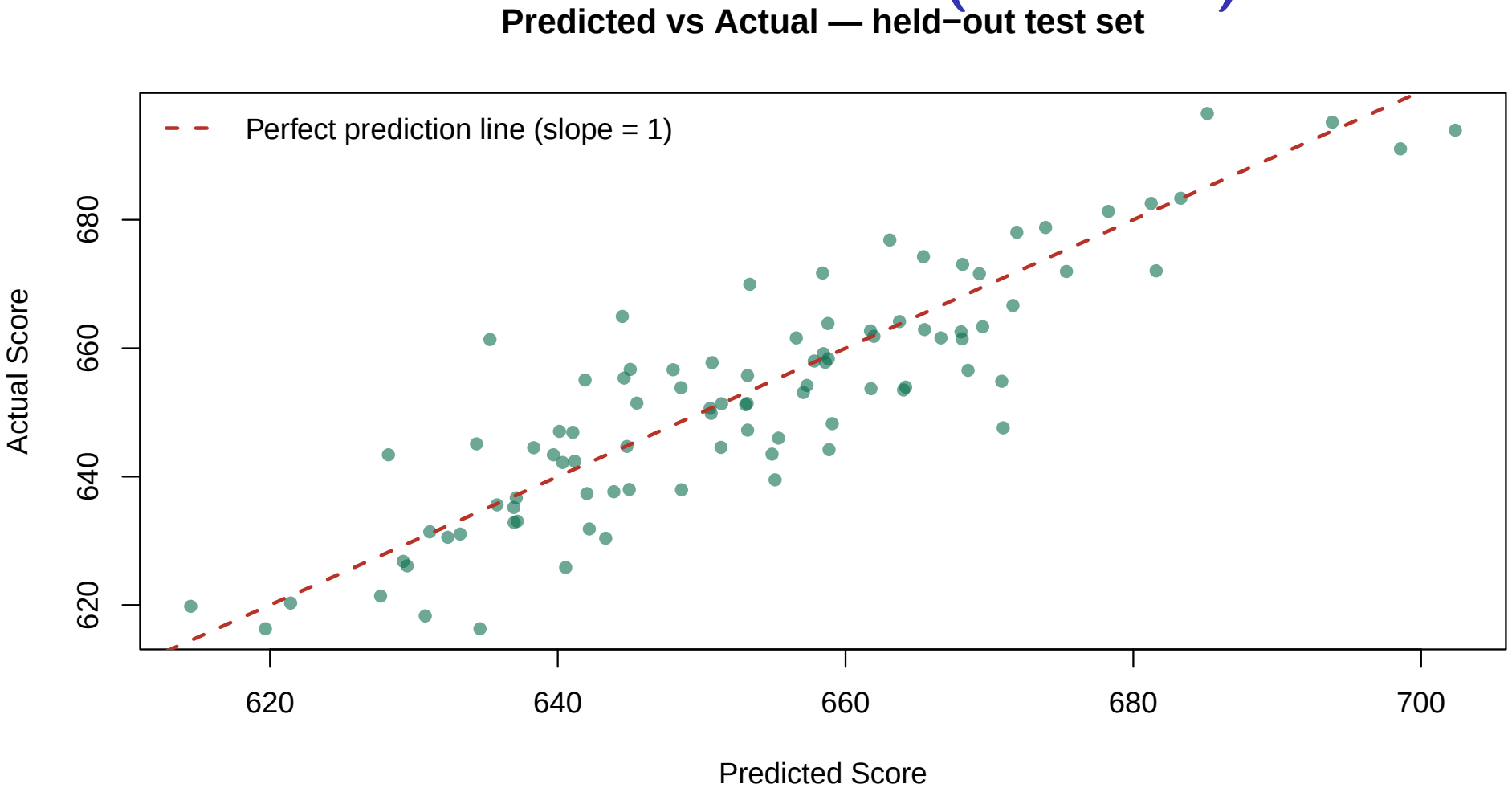
CASchools: Predicted vs Actual (Test Set)



- The tight scatter around the diagonal shows the model generalizes well to unseen districts.
- Residual spread is roughly even across score levels, consistent with the Breusch-Pagan result.

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Section 2

Part II

Finance Application

6 NYSE stocks, 3 years of close prices, and S&P 500 benchmarking

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Finance: Method Steps

Step	Detail
Pull close prices	Yahoo Finance, 2023-2025
Convert to daily returns	Arithmetic daily returns
Annualize return and risk	Mean*252 and sd*sqrt(252)
Compare correlation to market	Corr(stock, SP500)
Plot cumulative returns	Growth of $1 over time
Run CAPM regressions	alpha, beta, and R2
Track rolling beta	60-day beta windows
Track rolling correlation	60-day correlation windows

- Returns, not raw prices, are the right object for comparison because the stocks trade at very different price levels.
- The finance results answer different questions: average performance, market linkage, regression exposure, and time variation.

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Finance: Stock Selection

Ticker	Company	Industry
JPM	JPMorgan Chase	Financial Services
JNJ	Johnson & Johnson	Healthcare
XOM	Exxon Mobil	Energy
WMT	Walmart	Consumer Defensive
CAT	Caterpillar	Industrials
BA	Boeing Company	Aerospace

- All six stocks are NYSE-listed.
- Industries are intentionally different to avoid choosing five similar firms.
- Benchmark: `^GSPC` for the S&P 500.
- The cross-sector mix gives the portfolio visibly different beta and correlation behavior instead of five nearly identical names.

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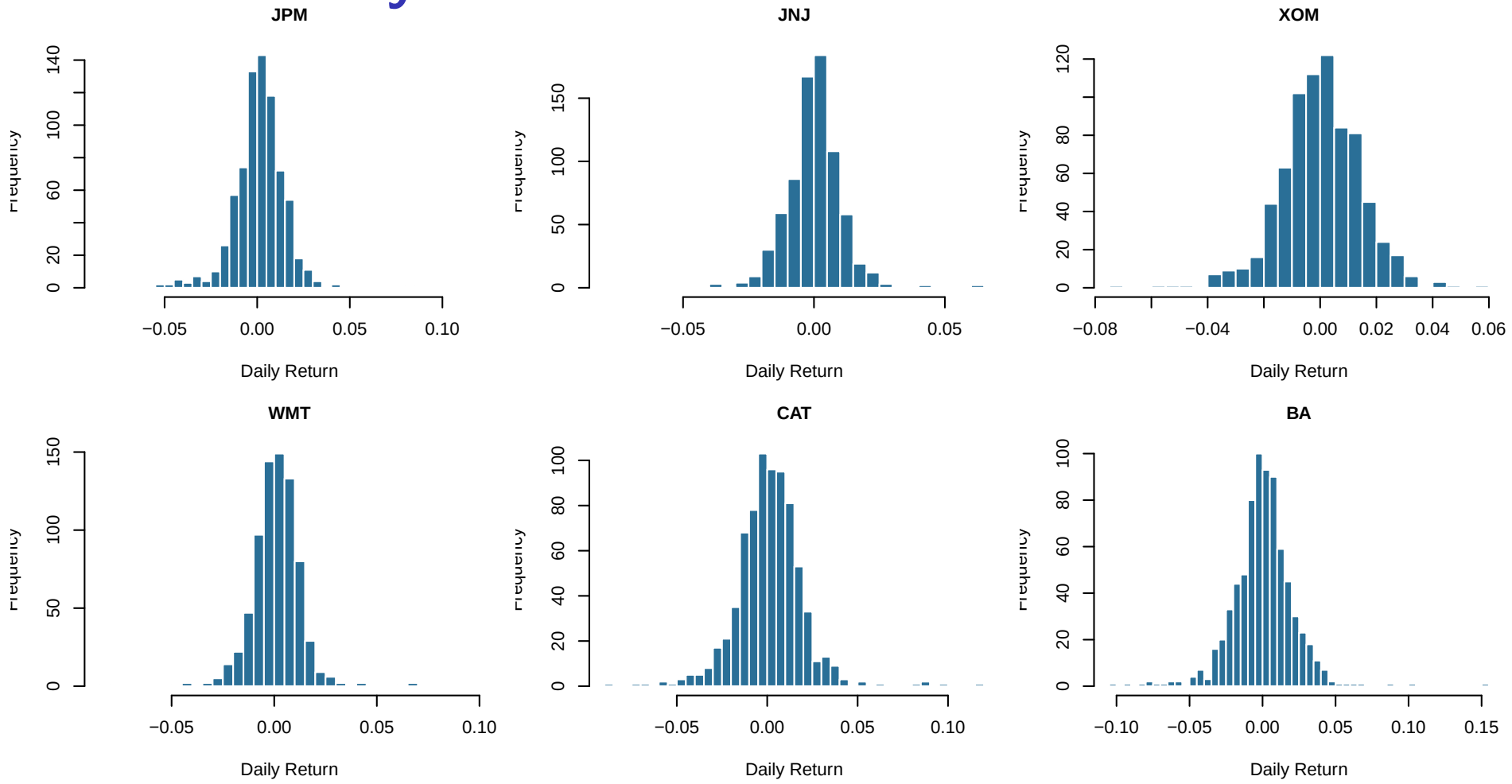
Finance: Daily Return Distributions



- All six stocks show fat-tailed return distributions — extreme days occur more often than a normal model predicts.
- The spread of each histogram confirms the annualized risk ordering shown in the return-risk plot.

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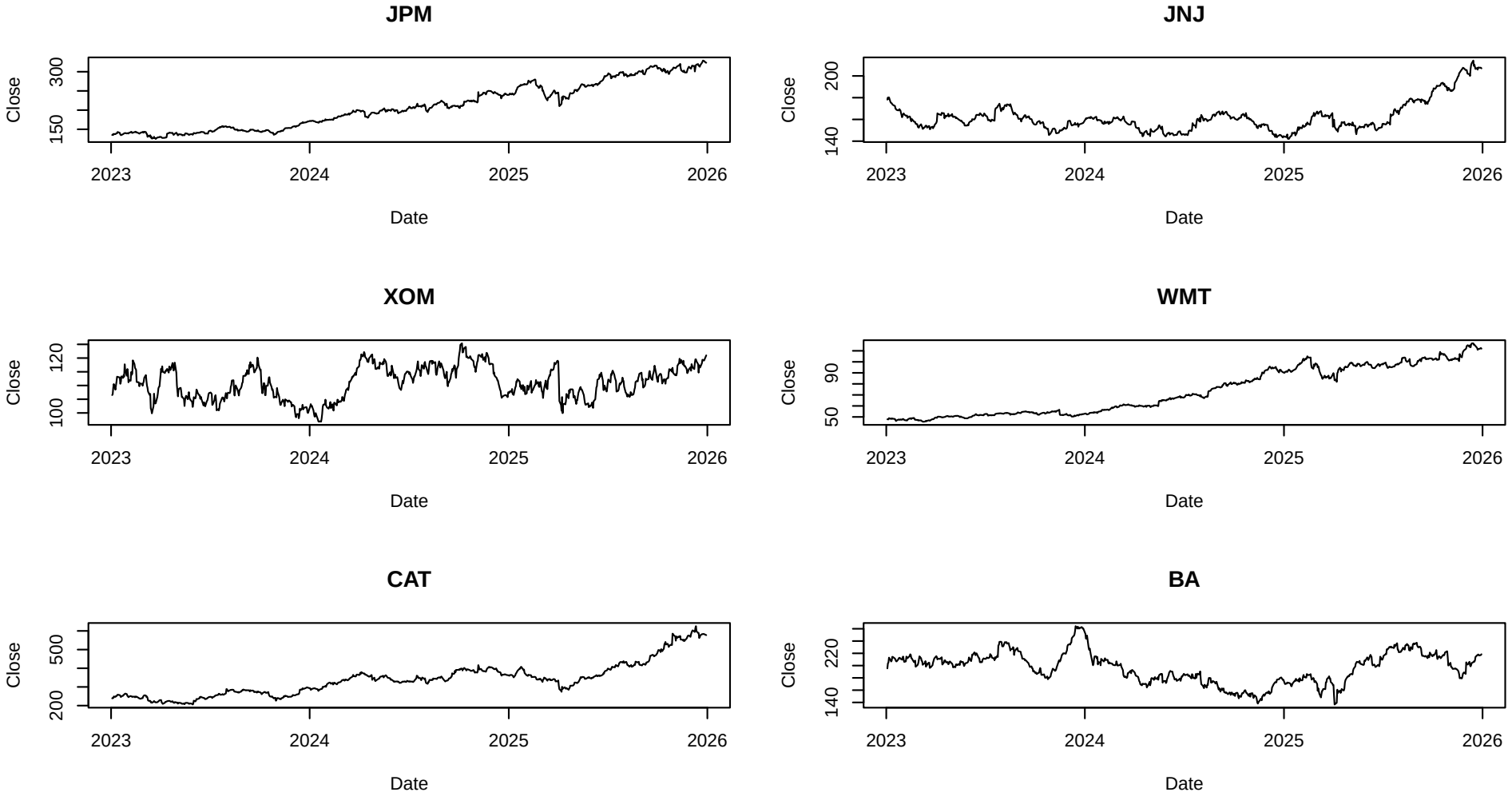
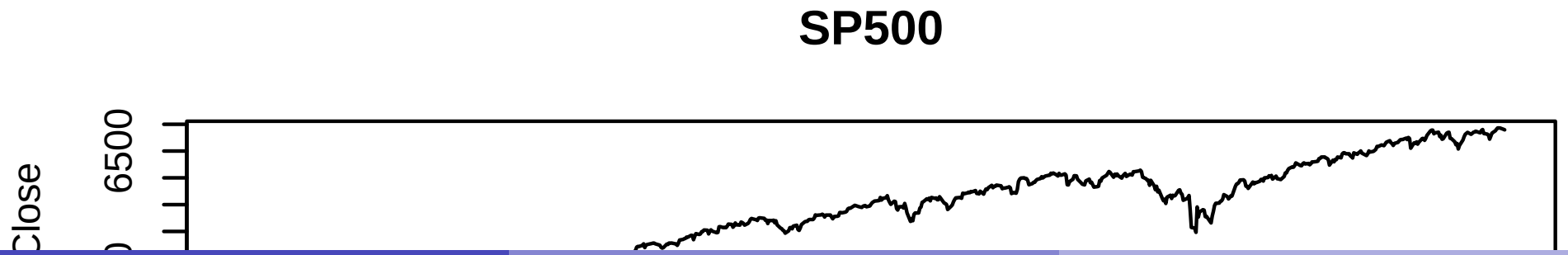
Finance: Close Prices





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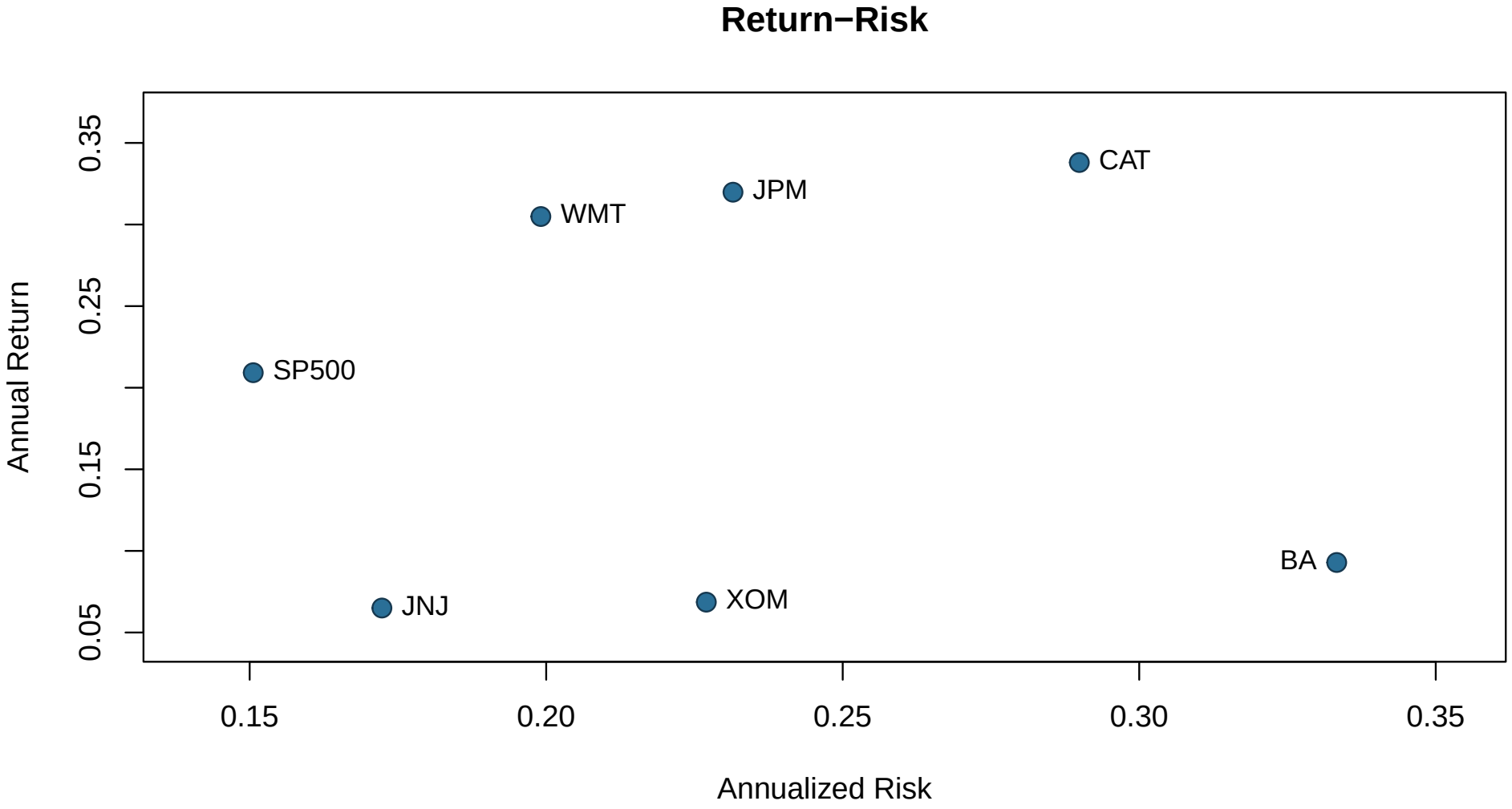
Finance: Annualized Return and Risk



- CAT has the strongest annualized return, while BA carries the most risk.
- JNJ is the defensive end of the group, so the chart separates aggressive from steadier names.

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Finance: Correlation with S&P 500 vs Risk and Return

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Ticker	Annual Return	Annual Risk	Corr with SP500
JPM	0.3198	0.2315	0.6023
JNJ	0.0650	0.1723	0.0811
XOM	0.0686	0.2270	0.2954
WMT	0.3049	0.1991	0.3494
CAT	0.3380	0.2899	0.6089
BA	0.0929	0.3333	0.4772

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Market correlation vs return

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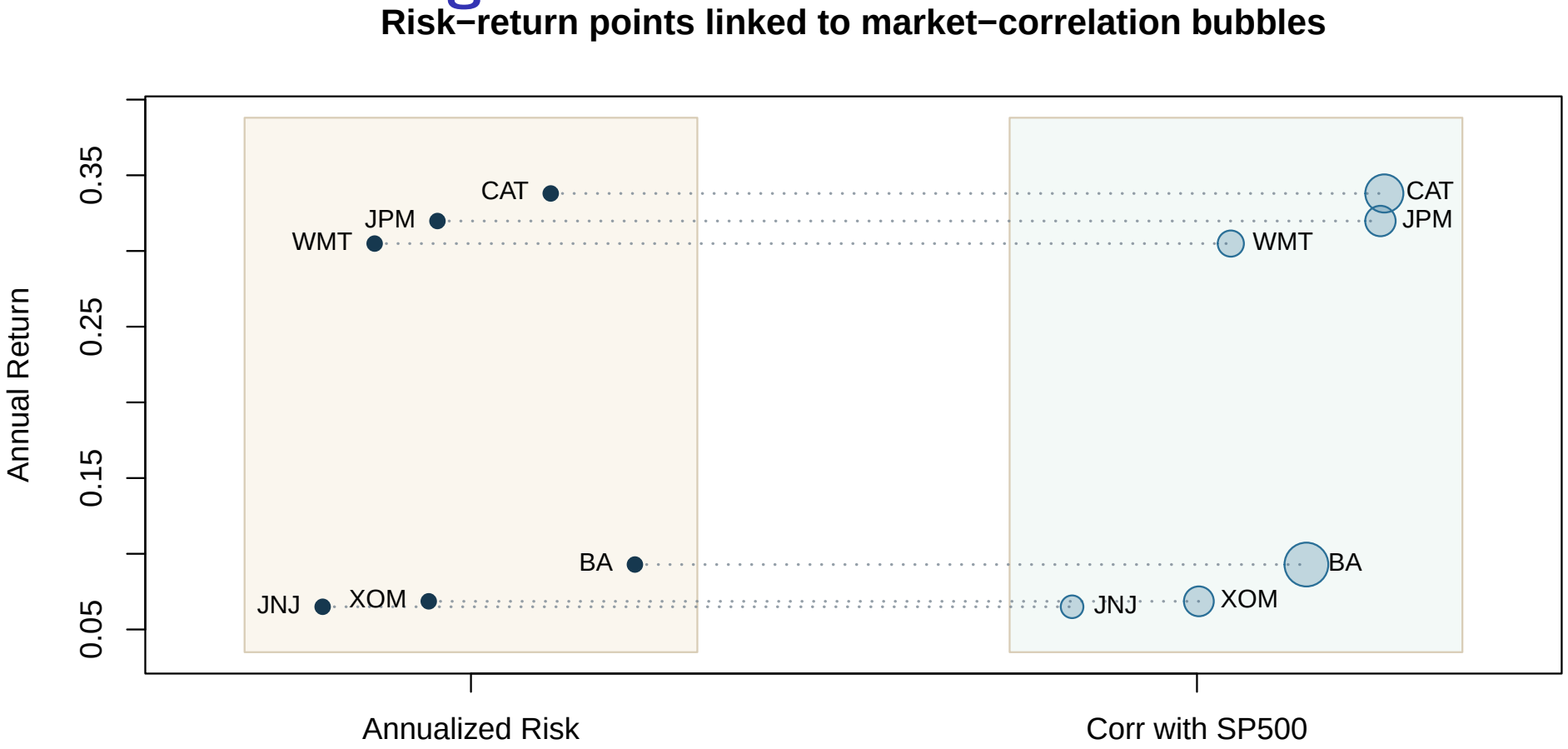
Finance: Linking Risk-Return and Market Correlation



- Dotted links connect each stock's annual return across the risk view and the market-correlation view.
- This makes it easier to compare whether a stock's return profile is lining up more with risk or with broad-market linkage.
- A stock can be very market-linked without having the most extreme

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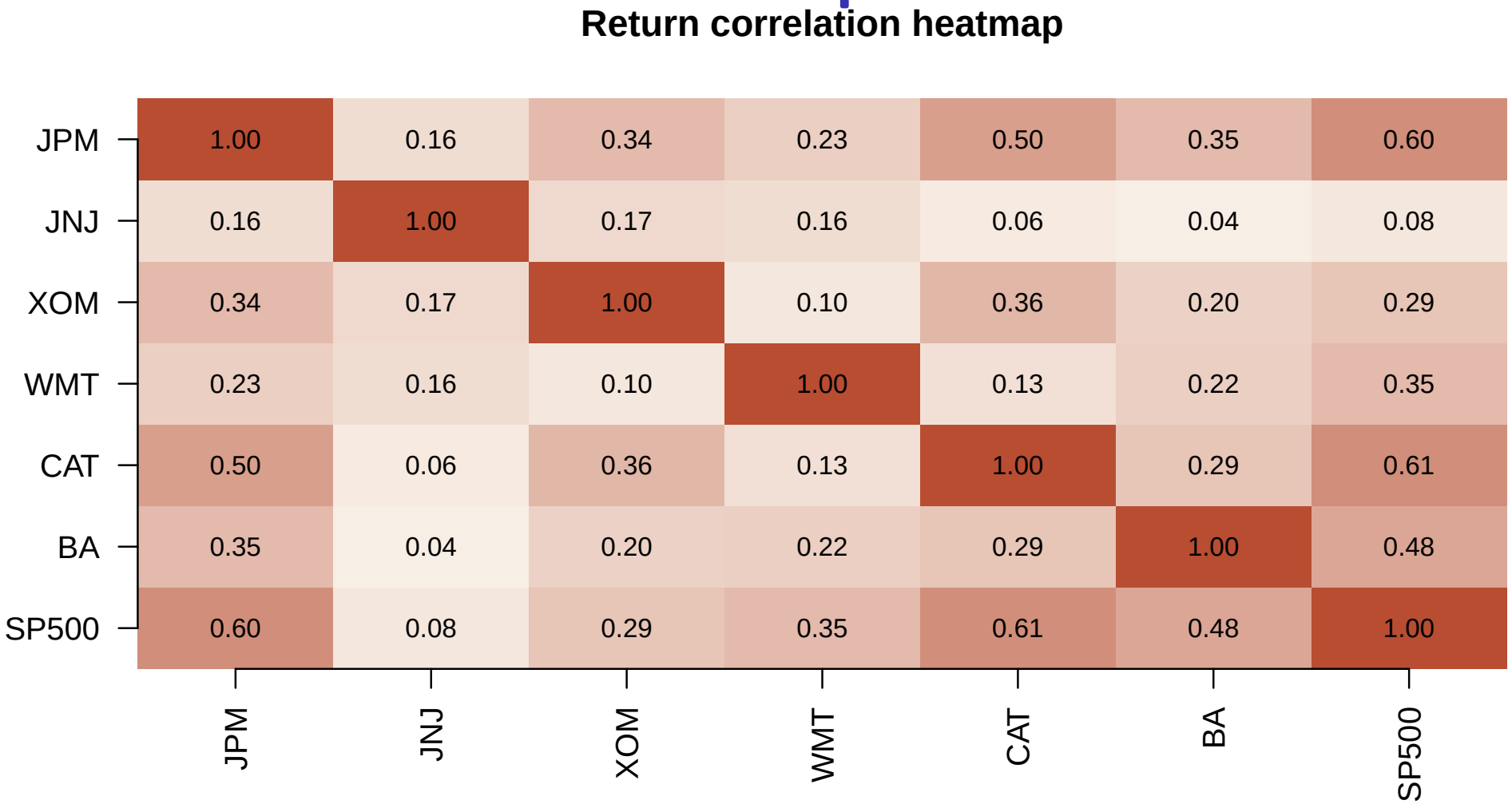
Finance: Correlation Heatmap



- Warmer colors mean stronger positive correlation.
- Cooler colors mean weaker market co-movement.
- Correlation matters because diversification is weaker when all assets move together.

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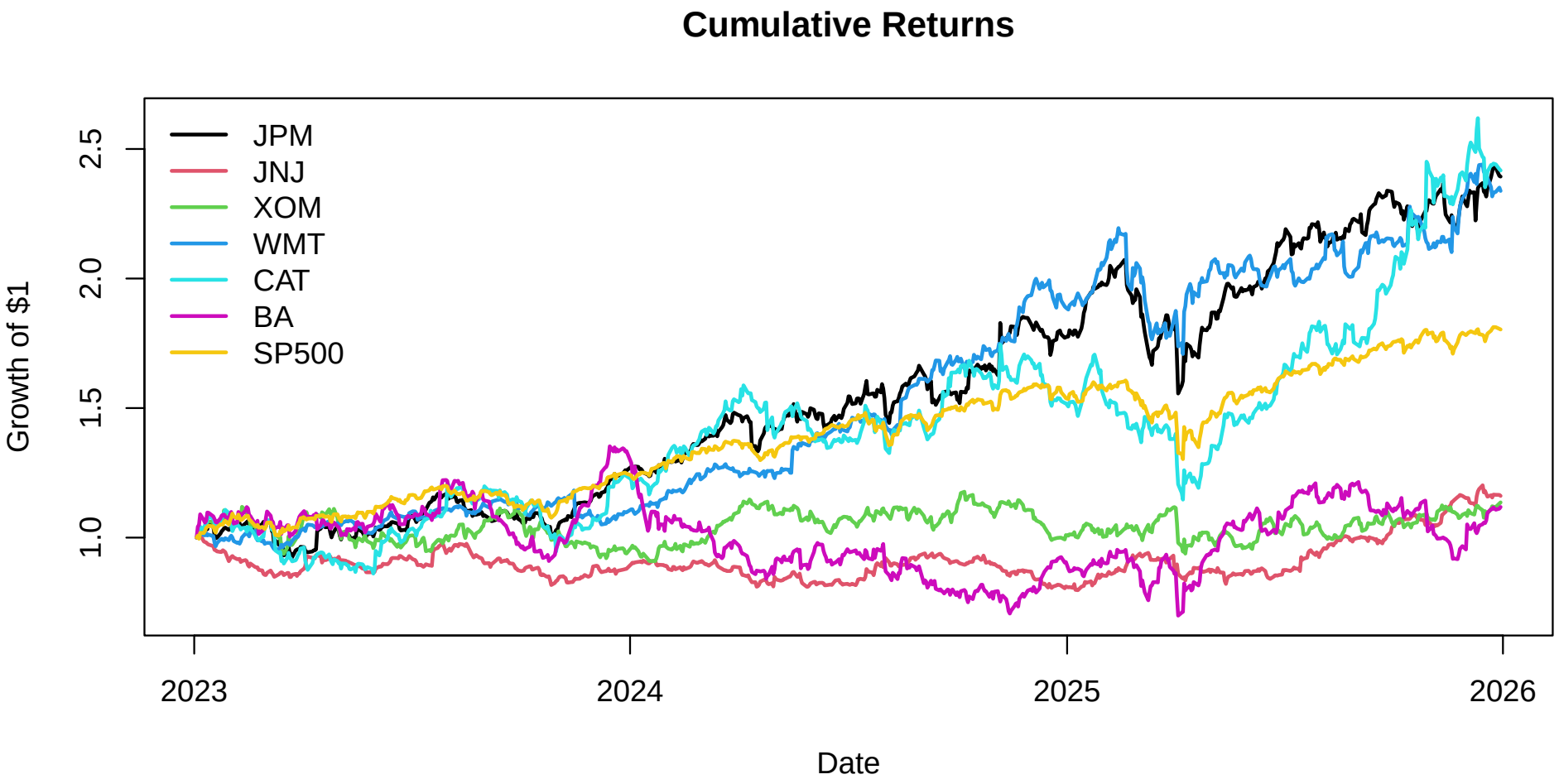
Finance: Cumulative Returns



- CAT finishes with the strongest growth path, while BA trails the group.
- The path differences show why average return alone can hide very different investment experiences.

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Finance: Alpha, Beta, and R-Squared

Ticker	Alpha Daily	Alpha Annual	Beta	R Squared
JPM	5e-04	0.1261	0.9257	0.3628
JNJ	2e-04	0.0456	0.0927	0.0066
XOM	-1e-04	-0.0246	0.4452	0.0873
WMT	8e-04	0.2083	0.4618	0.1221
CAT	4e-04	0.0929	1.1718	0.3707
BA	-5e-04	-0.1280	1.0558	0.2277

- `beta` measures sensitivity to market return.
- `alpha` measures abnormal return relative to the S&P 500.
- `R^2` measures how much of stock-return variation is explained by the market.
- CAT is the most market-sensitive name here, and CAT is the stock whose returns are most tightly tied to broad market movement.

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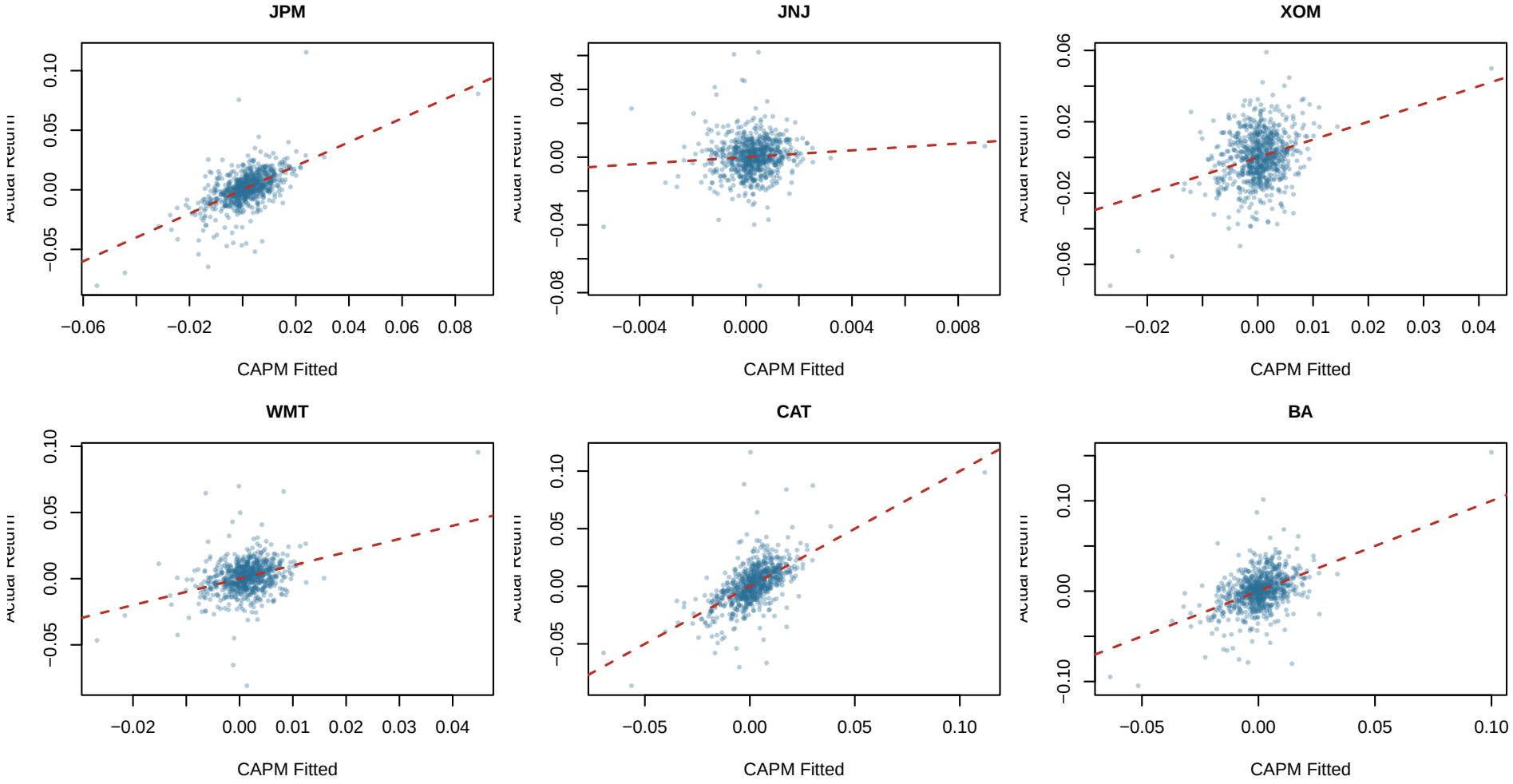
Finance: CAPM Fitted vs Actual Returns



- Stocks with higher R-squared cluster more tightly around the diagonal; lower R-squared stocks show wider scatter.
- The spread in each panel is the stock-specific variation the CAPM market factor cannot explain.

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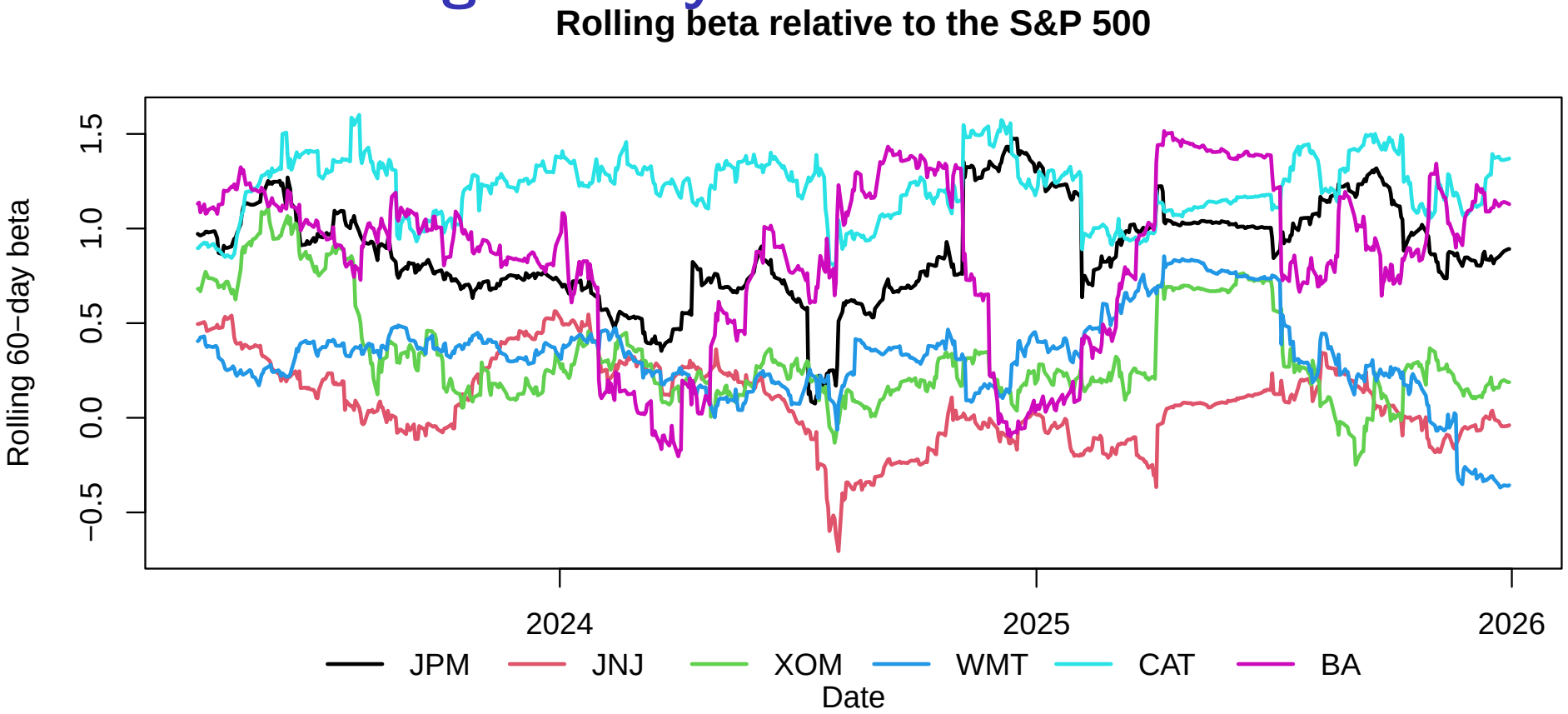
Finance: Rolling 60-Day Beta



- Beta above 1 means the stock moved more aggressively than the market in that 60-day window.
- Beta blends co-movement with relative volatility, so a rising line can mean tighter market linkage, larger stock swings, or both.
- A single full-sample beta misses these shifts, so rolling beta is more informative about changing exposure.

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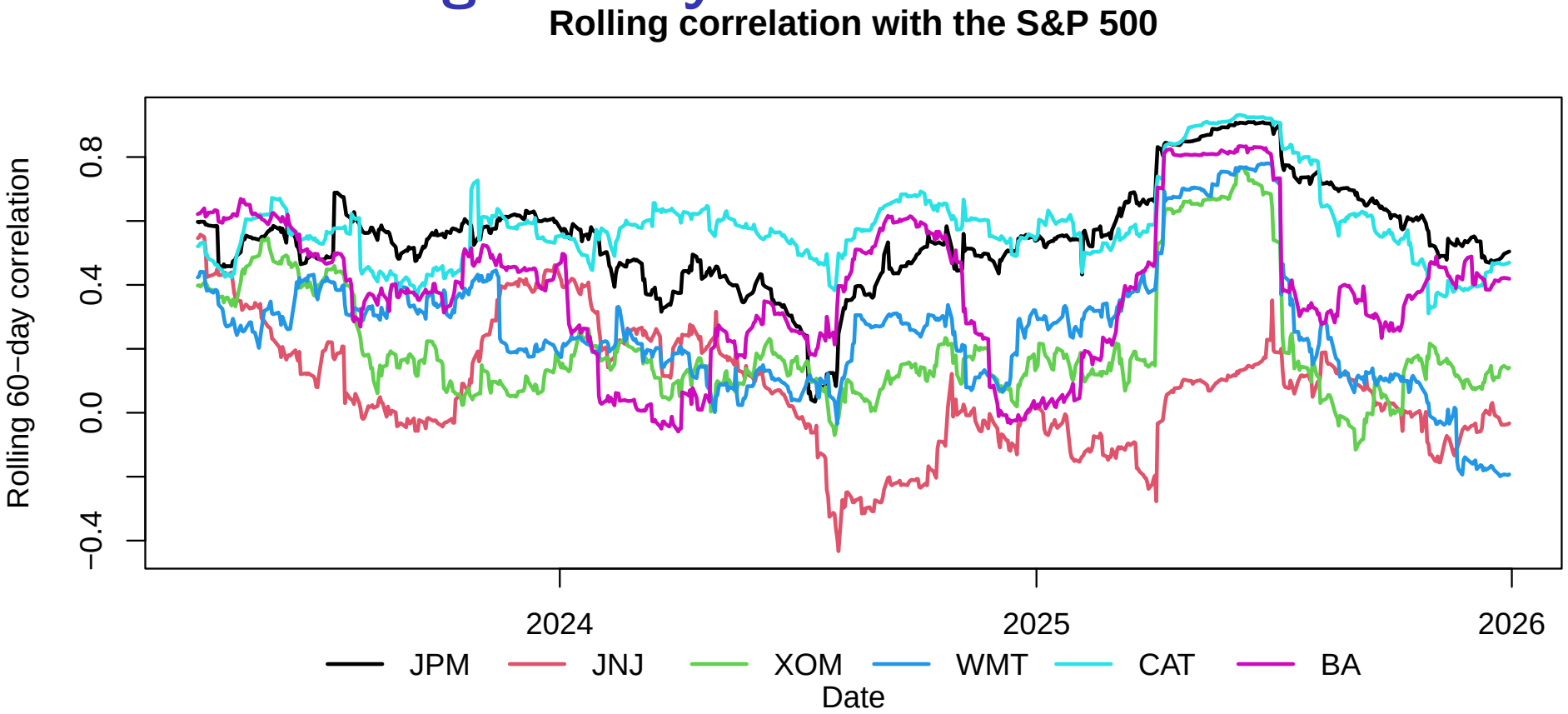
Finance: Rolling 60-Day Correlation



- Correlation isolates co-movement strength only, without the relative-volatility piece.
- Together, the rolling beta and rolling correlation plots show whether a stock became more market-linked because it moved more in sync with the S&P 500, because it got more volatile, or both.
- Comparing the two rolling plots separates "moves with the market"

class: center, middle, inverse

Section 3

Final Takeaways

Final Takeaways

- `HousePrices`: the best CV step is `S1`, and Cook filtering improves the whole validation ladder while keeping house-price prediction strong.
- `CASchools`: AIC and CV agree on the same reduced step, and the final model links stronger scores with lower poverty measures and higher district income.
- Finance: the six stocks occupy different risk-return positions, show different market linkage to the S&P 500, and that linkage changes over time in the rolling beta and rolling correlation plots.

class: center, middle, inverse

Section 4

Appendix

Appendix

CASchools district-type extensions

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Appendix: CASchools Within-Group Fit Check

District Type	N	Pooled Within-Group R2	Stratified Within-Group R2	R2 Gain from Splitting
KK-06	42	0.9201	0.9275	0.0074
KK-08	266	0.8432	0.8434	0.0002

- These are the right `R^2` comparisons to make: pooled-model fit inside each district type versus a separate model fit inside that same district type.
- Splitting helps `KK-06` by about 0.0074 and changes `KK-08` by only 0.0002.
- So the pooled model was already fitting both district types fairly well; stratifying is not a dramatic upgrade.

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Appendix: CASchools Stratified Coefficients and P Values

Term	KK-06 Est (p)	KK-08 Est (p)
str	-0.4693 (0.308301)	-0.9888 (0.000237)
calworks	0.0071 (0.964336)	0.0018 (0.977435)
lunch	-0.3527 (0.000905)	-0.4322 ($<$ 1e-16)
income	0.9616 (1.27e-05)	0.5758 (7.26e-09)
english	-0.2151 (0.030894)	-0.1597 (2.49e-05)

- `lunch` and `income` are stable in both district types.
- `english` is clearly significant in `KK-08`, but only marginal in the smaller `KK-06` sample.
- `str` stays negative in both groups, but its significance is much less stable than the socioeconomic measures.

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Appendix: CASchools Interaction Slopes by District Type

Model	Residual.DF	RSS	F	P.Value
Main-effects pooled model	302	15019.11		
Add 5 district-type interactions	296	14902.37	0.3864	0.887

Term	KK-06 Slope (p)	KK-08 Slope (p)	Delta 08-06 (p)
str	-0.4693 (0.444)	-0.9888 (0.000149)	-0.5195 (0.435)
calworks	0.0071 (0.973)	0.0018 (0.977)	-0.0053 (0.981)
lunch	-0.3527 (0.00767)	-0.4322 (<1e-16)	-0.0795 (0.563)
income	0.9616 (0.000212)	0.5758 (2.3e-09)	-0.3858 (0.158)
english	-0.2151 (0.0965)	-0.1597 (1.36e-05)	0.0555 (0.679)

- In R's treatment coding, `KK-06` is the baseline group, so the raw interaction coefficients are slope differences rather than full `KK-08` slopes.

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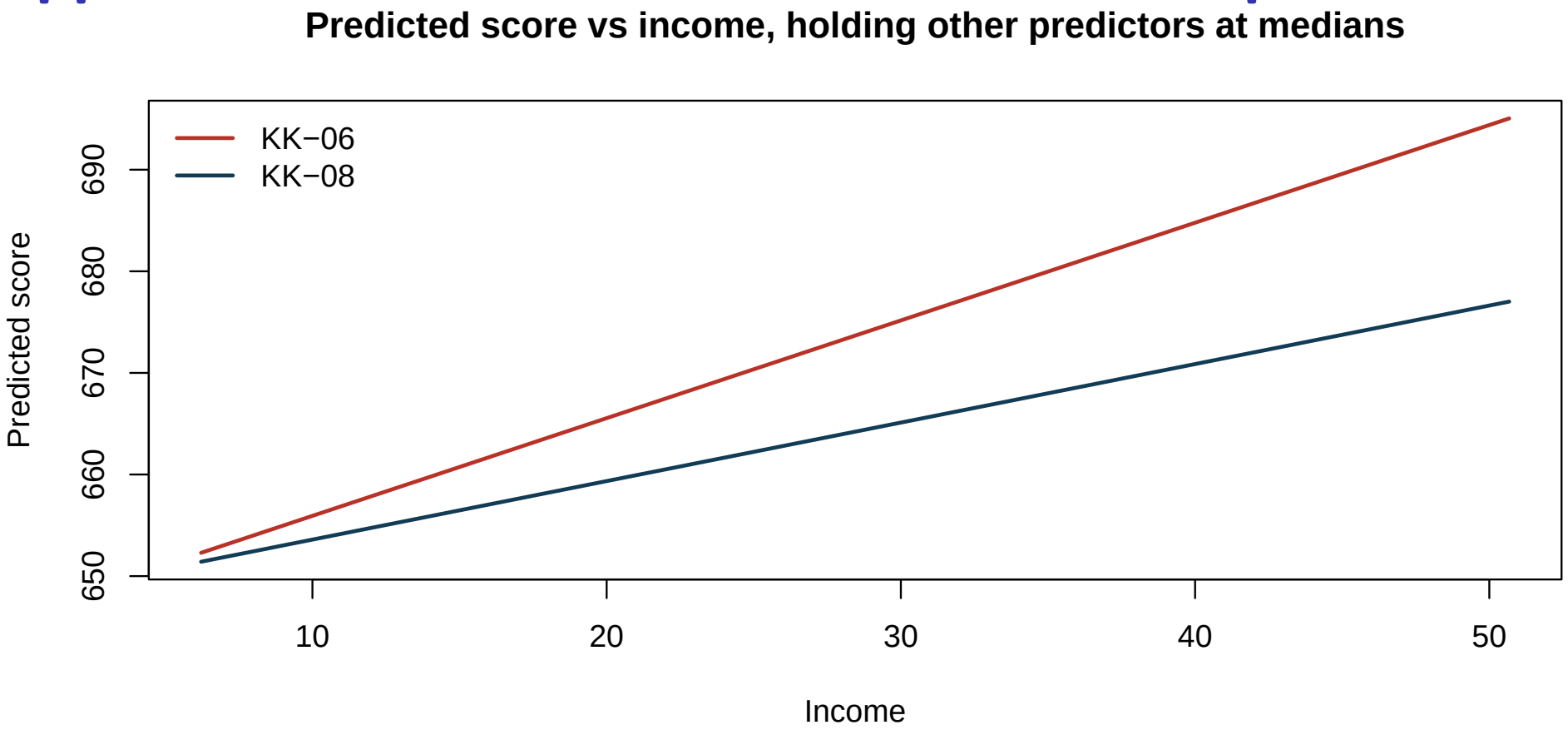
Appendix: CASchools Interaction Plot Explained



- The joint F-test gives a p-value of 0.887, so adding all five interactions does not materially improve fit.
- Only `income` varies here; the other predictors are fixed at their sample medians, and the near-parallel lines show district type mainly shifts the fitted level rather than the slope pattern.